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Competitive ability, body size and geographical range size in small mammals

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Abstract

Aim Why are some species widespread and abundant while others are restricted and rare? Darwin (1859) and others have claimed that some species have become widespread because they are competitively dominant over related, geographically restricted species. An alternative hypothesis is that wide-ranging species are ecological generalists that have been able to opportunistically colonize many kinds of new, disturbed and/or marginal habitats, whereas related narrow-ranging species are ecological specialists that competitively dominate specific kinds of relatively stable habitats/resources. We tested these opposing hypotheses using small mammals, for which considerable data on competitive interactions and geographical range sizes are available.

Location North America, Europe, Asia, Africa and Australia.

Methods We analysed data for forty-three competitive interactions between congeneric species in twenty genera, seven families, and four orders of small mammals. Competitive dominance was identified from both field and laboratory experiments, which were lumped because they showed similar results.

Results We found that (1) when all studies were analysed as individual data points, species with smaller geographical ranges tended to be dominant over congeneric species with larger geographical ranges, (2) lumping of interaction pairs sharing one or more of the same species yielded a similar result, (3) lumping all interactions involving the same genus also showed the same result, although it was not statistically significant, probably because of the small sample size resulting from this attempt to remove phylogenetic effects. Examination of the taxonomic relationships of the studied genera revealed no obvious phylogenetic effects on the relationship of competitive dominance with geographical range size. Furthermore, although body-size differences appeared to have played a role in the results observed, they cannot completely explain them.

Main conclusions We tentatively conclude that Darwin's hypothesis is falsified for small mammals, while the alternative hypothesis that geographically restricted species are competitively dominant over related widespread species receives some support.

Keywords
behavioural dominance, competition, congeneric species, geographical range size, mammals, meta-analysis.

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INTRODUCTION

Nearly 150 years ago, Darwin (1859) lamented in his book *On the Origin of Species*: 'Who can explain why one species ranges widely and is very numerous, and why another species has a narrow range and is rare? Yet these relations are of the highest importance, for they determine the present welfare, and, as I believe, the future success and modification of every inhabitant of this world' (p. 6). Despite this inspirational call for further research in one of the most famous scientific books of all time, we are still a long way from answering Darwin's fundamental question.

One approach to this problem, which has received increasing attention during the last two decades, is to search for biological correlates of geographical range size (see reviews of Kunin & Gaston, 1993; Gaston, 1994; Kunin & Gaston, 1997). Some factors that have been identified as being at least potentially important within specific taxa include dispersal/colonization ability (e.g. Glazier, 1980; Reaka, 1980; Oakwood *et al*., 1993; Kelly & Woodward, 1996; Gutiérrez & Menéndez, 1997; Dennis *et al*., 2000), niche breadth (e.g. Glazier, 1980; Brown, 1984, 1995; Novotný, 1991; Pagel *et al*., 1991; Kitahara & Fujii, 1994; Hecnar, 1999; Pyron, 1999; Dennis *et al*., 2000; Krüger & McGavin, 2000), trophic level/feeding habits (e.g. Brown, 1981; Rapoport, 1982; Pagel *et al*., 1991; Arita *et al*., 1997), body size (e.g. Glazier, 1980; Reaka, 1980; Brown, 1981, 1995; Aizen & Patterson, 1990; Arita *et al*., 1997; Gutiérrez & Menéndez, 1997; Hecnar, 1999; Pyron, 1999; Krüger & McGavin, 2000), life form (e.g. Oakwood *et al*., 1993; Kelly & Woodward, 1996), genetic variation (e.g. Karron, 1997), and life-history traits (e.g. Glazier, 1980, 1986; Aizen & Patterson, 1990; Oakwood *et al*., 1993; Dennis *et al*., 2000).

However, we are unaware of any comparative studies designed to test whether geographical range size is related to competitive ability. This is surprising because two prominent contrasting views on the competitive ability of widely vs. narrowly distributed species are present in the ecological and biogeographical literature. On one hand, Darwin (1859), and many others after him (e.g. Brown, 1957; Darlington, 1957; Hanski, 1982; Brown, 1984, 1995; Bowers, 1988), have argued that widespread species and higher taxa have superior 'fitness', and thus have attained their large ranges by out-competing restricted, 'less fit' taxa for geographical territory. The common finding that widespread species are usually more abundant than related restricted species in local habitats has been offered in support of this view (Brown, 1984, 1995).

On the other hand, some biologists have suggested that broadly distributed species may be competitively inferior to related, more narrowly distributed species. According to this view, the greater ecological specialization of restricted species has given them enhanced competitive ability. Wide-ranging species are thought to be opportunistic colonists of new, disturbed and/or marginal habitats scattered over wide areas, and as such are replaced by more ecologically specialized, restricted species in relatively stable habitats (e.g. Wilson, 1961; Ricklefs & Cox, 1972, 1978; Glazier, 1980, 1986, 1987; Novotný, 1991; Thomas, 1991; Dennis *et al*., 2000; references therein). This view is supported by two well-known models: the r- vs. K-selection model (MacArthur & Wilson, 1967) and the included niche model (Miller, 1967; Morse, 1974, 1980; Henttonen & Hansson, 1984). According to the r- vs. K-selection model, species that use relatively unstable habitats or resources are selected for high colonizing ability and rapid population growth, whereas species that use relatively stable habitats or resources are selected for high competitive ability and efficient use of resources. The included niche model predicts that ecologically specialized species should out-compete coexisting generalized species, which are consequently forced to occupy marginal niche space. Both of these models can explain why geographically restricted species may be competitively dominant over related widespread species.

Of course, a third possibility is that geographical range size is unrelated to competitive ability. This may be because other biological, historical, or environmental factors are more important, or because relative competitive ability is so habitat- or taxon-specific that no consistent regional differences between wide- vs. narrow-ranging species can be discerned.

These alternative hypotheses were tested by comparing the relative competitive abilities of congeneric species of small mammals with different geographical range sizes. Small mammals were chosen for this study because their geographical range sizes are well known, and considerable data on interspecific competitive interactions are available. Only pairs of congeneric species were studied to maximize the likelihood of observing competitive effects and to minimize extraneous phylogenetic effects.

METHODS

We analysed data for forty-three competitive interactions between congeneric species in twenty genera, seven families and four orders of small mammals (Appendix 1). Three kinds of studies were included and initially analysed separately: laboratory studies, field studies, and studies that combined laboratory and field experiments.

Different investigators had different specific definitions of competitive dominance, but in all studies a dominant species was regarded as having a greater adverse effect on a subordinate species, than the reverse. The laboratory studies typically examined the relative frequency of agonistic and submissive behaviours in staged dyadic encounters, the more aggressive species being considered dominant (e.g. Sheppard, 1971; Blaustein & Risser, 1976; Montgomery, 1978; Dempster & Perrin, 1990; Livoreil *et al*., 1993). This approach was a reliable way to measure aggressiveness (i.e. interference competition), because it is robust to variation in testing protocols (Harper & Batzli, 1997). Most of the field studies either made direct observations of relative behavioural dominance in natural areas (e.g. Chappell, 1978) baited areas (e.g. Congdon, 1974), or outdoor pens or cages (e.g. Hoffmeyer, 1973; Wolff *et al*., 1983; Fitzgerald & Keith, 1990), or inferred competitive dominance from

estimates of relative abundance, relative survival, home-range size, foraging success, reproductive output, activity patterns, and/or microhabitat distribution in study plots ranging in size from 0.00027 to 3 km^2. The latter field studies involved spatial or temporal comparisons of relative abundance or other indices of competitive success in unmanipulated plots (e.g. Tast, 1968; Hallett, 1982; Viitala, 1984; Norrdahl & Korpimäki, 1993) or plots or enclosures where competing species had been introduced (e.g. Vaughan & Hansen, 1964; Ziv *et al*., 1993), removed (e.g. Chappell, 1978; Holbrook, 1979; Gliwicz, 1981), or both (Dickman, 1986). Two field studies inferred competitive dominance from regional analyses of apparent competitive exclusion among species (Kennerly, 1959; Miller, 1964). Results were lumped for multiple studies examining competitive interactions between the same two species. If the results were conflicting they were resolved, as explained in Appendix 2.

Relative geographical range sizes (larger vs. smaller) were determined using Hall (1981) and Nowak (1999). Four interaction pairs (*Rattus rattus* L./*R. norvegicus* Berkenhout, *Tamias dorsalis* Baird/*T. umbrinus* J.A. Allen, *Clethrionomys gapperi* Vigors/*C. rutilus* Pallas and *Ctenomys talarum* Thomas/*C. australis* Rusconi) were excluded from the analyses because the geographical range sizes of the species in each pair were virtually the same (Barnett & Spencer, 1951; Brown, 1971; McPhee, 1984; Comparatore *et al*., 1992; Vassallo, 1993). Four other interaction pairs (*Microtus pennsylvanicus* Ord/*M. ochrogaster* Wagner, *M. pennsylvanicus*/*M. montanus* Peale, *Peromyscus maniculatus* Wagner/*P. truei* Shufeldt and *Sylvilagus floridanus* J.A. Allen/*S. transitionalis* Bangs) were also omitted because no dominance hierarchy could be demonstrated (Getz, 1962; Koplin & Hoffman, 1968; Miller, 1969; Murie, 1971; Grant, 1972; Stoecker, 1972; Krebs, 1977; Llewellyn, 1980; Probert & Litvaitis, 1996).

Body size was examined because it can contribute to competitive dominance (Miller, 1967; Grant, 1972; Morse, 1974; Viitala, 1977; Henttonen & Hansson, 1984; Persson, 1985). Relative body masses (larger vs. smaller) were determined using data in Silva & Downing (1995).

G-tests of independence (using Williams's correction, following Sokal & Rohlf, 1995) were used to test for relationships between competitive dominance and geographical range size, between competitive dominance and body mass, and between geographical range size and body mass. The tests were made in three ways. First, each interaction pair was treated as a separate data point. Secondly, interaction pairs involving the same species were lumped and counted as a single data point. Thirdly, interaction pairs involving the same genus were lumped and counted as a single data point. In this lumping process, when studies did not agree the prevailing trend was used. If there was no prevailing trend, the data were omitted. This rarely occurred: once when examining competitive dominance in relation to geographical range size, once when examining competitive dominance in relation to body mass, and thrice when examining geographical range size in relation to body mass.

The last two methods were employed to minimize phylogenetic bias (Harvey & Pagel, 1991). An exploration of possible phylogenetic effects was also made by comparing results in different taxa using the taxonomic scheme of Nowak (1999).

RESULTS

The forty-three interaction pairs used in this study, and their taxonomic affinities, are summarized in Appendix 1. Results of field and laboratory studies were similar: relatively restricted species were more often competitively dominant over congeneric widespread species than vice versa (Table 1). These results were significant for field studies and for all studies combined, and marginally significant for laboratory studies. As the field and laboratory studies revealed similar results they were analysed together in the rest of this section. The results were also not affected by the fact that in some interaction pairs the species with a smaller distribution was still quite wide-ranging (e.g. *Sorex araneus* L., *Lepus europaeus* Pallas and *P. leucopus* Rafinesque).

Competitive dominance and geographical range size were significantly inversely associated for all interaction pairs, including when interaction pairs sharing the same species were lumped (Table 2). A similar result was observed when all interaction pairs in the same genus were lumped, although it was not significant, probably because of the small sample size resulting from this lumping process (Table 2). Examination of the taxonomic relationships of the studied genera revealed no obvious phylogenetic effects on the relationship of competitive dominance with geographical range size, although this conclusion should be re-examined when more data on non-rodent competitive

Table 1 Comparison of study types, counting all interaction pairs of congeneric species of small mammals as individual data points

Study type	Dominant has larger geographical range	Dominant has smaller geographical range	χ^2
Laboratory studies	4	11	3.29*
Field studies	4	13	4.76**
Both field & laboratory studies for each interaction pair	4	7	0.08
All studies combined	12	31	8.59****

Chi-square values (χ^2) for associations between competitive dominance and relative geographical range size are shown with significance levels (*$P < 0.10$; **$P < 0.05$; ***$P < 0.01$; ****$P < 0.005$; *****$P < 0.001$).

Table 2 Effects of different data treatments on associations between competitive dominance and relative geographical range size among congeneric species pairs of small mammals

Study type	Dominant has larger geographical range	Dominant has smaller geographical range	χ^2
All pairs as individual data points	12	31	8.59****
Lumping of pairs involving same species	8	22	6.68***
Lumping of pairs in same genus	6	13	2.57

Chi-square values (χ^2) with significance levels are shown (*P < 0.10; **P < 0.05; ***P < 0.01; ****P < 0.005; *****P < 0.001).

interactions become available (see Appendix 1). Most (36) of the interactions observed in this study were between rodents. Nevertheless, for both rodent and non-rodent interactions, species with the smaller geographical ranges were more often competitively dominant than species with larger ranges (27 vs. nine, and four vs. three, respectively).

Competitive dominance was highly significantly associated with larger body mass, as shown by all three methods of comparison (Table 3). Geographical range size also tended to be inversely related to body mass (Table 4). However, the latter tendency was only marginally significant when all interaction pairs were treated as individual data points, and it lost significance when interaction pairs with the same species were lumped, or when interaction pairs in the same genus were lumped.

DISCUSSION

This study tested whether geographical range size is related to the relative competitive ability of congeneric species of small mammals. There are three possibilities: (1) widespread species are competitively dominant over relatively restricted species, (2) restricted species are dominant over relatively widespread species, or (3) there is no association between geographical range size and competitive ability. The present analysis suggests that the second alternative is the most important one. Using three methods of data analysis, competitive dominance was found to be more prevalent among geographically restricted species than among related widespread species (Tables 1 and 2).

According to Glazier (1987), restricted species have tended to evolve in relatively stable habitats where K-selection for high competitive ability is favoured. This hypothesis is supported by the finding that restricted species of rodents and moths tend to have more stable populations than related widespread species (Glazier, 1986). It is further supported by an extensive comparative analysis of the life-history traits and habitat distribution of *Peromyscus* species (Glazier, 1980). Widely distributed species in this genus tend to occur more frequently in new, disturbed and/or marginal habitats than narrowly distributed species. Moreover, the habitat distribution of the widespread species tends to

Table 3 Effects of different data treatments on associations between competitive dominance and relative body mass among congeneric species pairs of small mammals

Study type	Dominant has larger body mass	Dominant has smaller body mass	χ^2
All pairs as individual data points	30	5	19.53*****
Lumping of pairs involving same species	23	2	14.88*****
Lumping of pairs in same genus	15	2	10.93*****

Chi-square values (χ^2) with significance levels are shown (*P < 0.10; **P < 0.05; ***P < 0.01; ****P < 0.005; *****P < 0.001).

Table 4 Effects of different data treatments on associations between relative body mass and relative geographical range size among congeneric species pairs of small mammals

Study type	Larger body mass with larger geographical range	Smaller body mass with larger geographical range	χ^2
All pairs as individual data points	12	23	3.47*
Lumping of pairs involving same species	8	16	2.66
Lumping of pairs in same genus	5	10	1.65

Chi-square values (χ^2) with significance levels are shown (*P < 0.10; **P < 0.05; ***P < 0.01; ****P < 0.005; *****P < 0.001).

shrink at localities with coexisting restricted species, as would be expected if the restricted species were competitively dominant. In addition, the restricted species have significantly lower reproductive potentials, as predicted from K-selection operating in relatively stable populations.

What gives many restricted small mammal species a competitive advantage over their more wide-ranging relatives? Ecological specialization, a common trait of narrowly distributed species (e.g. Glazier, 1980), is probably involved. Greater ecological specialization may permit a more efficient use of resources (i.e. greater exploitative competition), or may favour the evolution of greater aggressiveness (i.e. interference competition) as a way of protecting specialized niches from invasion by generalized species (Miller, 1967; Colwell & Fuentes, 1975). Many studies have shown that specialized species tend to have higher competitive ability than related generalized species (e.g. Miller, 1967; Morse, 1974; Holbrook, 1978; Hallett, 1982). Further research is needed to identify the precise mechanisms involved and their relative importance.

The results of this study are consistent with two additional general observations. First, most taxa include far more restricted species than widespread species (e.g. Rapoport, 1982; Gaston, 1994; Brown, 1995). If widespread species were competitively superior, they should have over-run the habitats of most restricted species, causing their extinction. That they have not attests to the ability of restricted species to hold their own against widespread species (either through superior competitive ability in specific habitats, or by occupation of refugia protected by barriers).

Secondly, in most taxa, species with small ranges are usually more numerous in the tropics than at higher latitudes (e.g. Rapoport, 1982; Glazier, 1987; Stevens, 1989; Brown, 1995; Dynesius & Jansson, 2000; Gaston & Blackburn, 2000). If one assumes that interspecific competition is generally more severe in the species-rich tropics, then the survival there of legions of narrow-ranging species argues against the idea that they are inferior competitors.

The included niche model of Miller (1967) is consistent with the present results, as well. Miller (1967) and Morse (1974, 1980) have reviewed many avian and mammalian examples of specialized species having smaller niche spaces enclosed within the larger niche spaces of more generalized related species. In most of these cases, it is the specialized species that are competitively dominant. They also note that many of these specialized species have larger body sizes. Larger animals are almost always better at interference competition than smaller animals (Persson, 1985). This is also true for the small mammals included in this study. Large species are more often competitively dominant over related smaller species than vice versa (Table 3).

One may argue that the association between competitive dominance and geographical range size observed in this study is merely a secondary result of both being correlated with body size. However, although competitive dominance is significantly related to body size, geographical range size is only weakly related to body size (Table 4). Thus, although body-size differences appear to have contributed to the association between competitive dominance and geographical range size, they do not completely explain this pattern. Larger body size may very well be one of the several traits that restricted species have evolved to make themselves more effective competitors with relatively generalized, widespread species that are constantly threatening to invade their niche spaces.

The present results contradict the view of Darwin (1859), Brown (1984) and others that widespread generalized species are competitively dominant over restricted specialized species. Although Brown presented no data on relative competitive ability, he inferred that widespread species are competitively dominant over restricted species because usually they are more abundant in local habitats. According to Brown (1984, 1995), a trade-off between being specialized vs. generalized (jack of all trades, but master of none) may not exist because, if it operated, specialized restricted species should be more abundant locally than generalized widespread species, which is generally not true.

However, a positive association between local abundance and geographical range does not necessarily mean that widespread species are competitively dominant over related restricted species. Plausible counter-arguments can be made for both exploitative and interference competition. For example, specialized restricted species could be more efficient resource exploiters than generalized widespread species, and still have smaller local populations because they use rarer or lower quality resources. In addition, species selection among restricted, locally rare species may have put a high premium on effective interference competition because it ensures that the limited resources on which these species have specialized will not be usurped by relatively generalized species. Restricted species are on the edge of extinction, and thus those that have evolved behavioural dominance are more likely to have survived than those that did not.

Thus, unlike Brown (1984, 1995), we suggest that trade-offs between being specialized vs. generalized have been important in determining the different geographical range sizes of related species. Other trade-offs, such as those between competitive ability (enhanced in restricted species) and dispersal/colonization proficiency (enhanced in widespread species), may also be important. Such trade-offs have been assumed in many species-coexistence models (e.g. Levins & Culver, 1971; Horn & MacArthur, 1972; Hastings, 1980; Tilman *et al.*, 1997; Holmes & Wilson, 1998; Kinzig *et al.*, 1999), and have been invoked to explain several ecological and biogeographical patterns, including succession (Tilman, 1990, 1994), species packing in communities (Kinzig *et al.*, 1999), the taxon cycle (Wilson, 1961; Ricklefs & Cox, 1972; Glazier, 1980), effects of habitat destruction on species loss (Tilman, 1994; Tilman *et al.*, 1997), and the tendency of many widespread species to inhabit new, disturbed and/or marginal, species-poor habitats in contrast to related restricted species that preferentially occupy relatively stable, species-rich habitats (e.g. Glazier, 1980, 1986; Novotný, 1991; references therein). Considerable empirical evidence for colonization-competition trade-offs exists, as well (see Holmes & Wilson, 1998).

The generalist vs. specialist and colonization vs. competition trade-offs may act in concert with other trade-offs resulting from r- vs. K-selection to influence patterns of geographical range sizes among related species (see Glazier, 1980, 1986). In support, Bowers & Harris (1994) have developed a metapopulation model showing that changing environments favour generalists with good dispersal ability, whereas constant environments favour specialists with high efficiency of resource utilization. The model of Pulliam (2000) also shows how dispersal ability and niche breadth may be linked to determine species-distribution patterns. Kitahara & Fujii (1994) have further argued that a synthesis of the generalist vs. specialist and r- vs. K-selection dichotomies are needed to better understand the biodiversity and structure of natural communities. Thus, the above trade-offs may represent interrelated aspects of a syndrome of factors that cause some species to be more widespread than other species. Further exploration of how these trade-offs interact to affect the geographical spread of a species may take us a long way toward answering Darwin's fundamental question quoted at the beginning of this paper. However, if the present results are generally applicable, the answer may be quite different from what Darwin anticipated.

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Appendix 1 Summary of interaction pairs (D = dominant, SU = subordinate, L = larger, S = smaller)

Interaction pairs	Competitive dominance	Geographical range	Body mass	Source
Infraclass: Metatheria				
Superorder: *Marsupialia*				
Order: Dasyuromorphia				
Family Dasyuridae				
Antechinus swainsonii	D	S	L	Dickman (1986, 1991)
A. stuartii	SU	L	S	
Infraclass: Eutheria				
Order: *Insectivora*				
Family Soricidae				
Neomys fodiens	D	L	L	Krushinska *et al.* (1994)
N. anomalus	SU	S	S	
S. araneus	D	S	L	Dickman (1991)
S. minutus	SU	L	S	Disagrees: Ellenbroek (1980)
S. vagrans	D	L	L	Dalquest (1941)
S. trowbridgii	SU	S	S	
Order: *Lagomorpha*				
Family: Leporidae				
Lepus arcticus	D	S	L	Fitzgerald & Keith (1990)
L. americanus	SU	L	S	
L. europaeus	D	S	L	Wolfe *et al.* (1996)
L. timidus	SU	L	S	
Sylvilagus floridanus	D	L	L	Chapman & Verts (1969)
S. bachmani	SU	S	S	
Order: Rodentia				
Suborder: Sciuromorpha				
Family: Sciuridae				
Spermophilus mexicanus	D	S	L	Livoreil *et al.* (1993)
S. spilosoma	SU	L	S	
*Tamias amoenus**	D	S	L	Heller (1971), Sheppard (1971),
*T. minimus**	SU	L	S	Chappell (1978)
*T. speciosus**	D	S		Chappell (1978)
*T. amoenus**	SU	L		Disagrees: Heller (1971)
*T. speciosus**	D	S		Chappell (1978)
*T. minimus**	SU	L		
*T. alpinus**	D	S		Heller (1971)
*T. speciosus**	SU	L		
Suborder: Myomorpha				
Family: Geomyidae				
Geomys personatus	D	S		Kennerly (1959)
G. bursarius	SU	L		
Thomomys bottae	D	S	S	Miller (1964), Vaughan & Hansen (1964)
T. talpoides	SU	L	L	Disagrees: Baker (1974)
Family: Heteromyidae				
Chaetodipus penicillatus†	D	L		Hoover *et al.* (1977), Hallett (1982)
Chaetodipus intermedius†	SU	S		
Dipodomys ordii	D	L	L	Schroder & Rosenzweig (1975),
D. merriami	SU	S	S	Blaustein & Risser (1976)
D. panamintinus	D	S	L	Blaustein & Risser (1976)
D. merriami	SU	L	S	
D. panamintinus	D	S	L	Blaustein & Risser (1976)
D. ordii	SU	L	S	
D. agilis	D	S	L	Christopher (1973)
D. merriami	SU	L	S	
D. stephensi	D	S	L	Price *et al.* (1991)
D. agilis	SU	L	S	
D. deserti	D	S	L	Congdon (1974)

Appendix 1 *continued*

Interaction pairs	Competitive dominance	Geographical range	Body mass	Source
D. merriami	SU	L	S	
Family: Muridae				
Subfamily: Arvicolinae				
Clethrionomys rufocanus	D	S	L	Viitala (1984)
C. rutilus	SU	L	S	
Microtus pinetorum	D	S	S	Novak & Getz (1969),
M. pennsylvanicus	SU	L	L	Cranford & Derting (1983)
M. oeconomus	D	L	L	Tast (1968), Henttonen *et al.* (1977)
M. agrestis	SU	S	S	
M. epiroticus‡	D	S	S	Myllymäki (1977),
M. agrestis	SU	L	L	Norrdahl & Korpimäki (1993)
Subfamily: Gerbillinae				
Gerbillurus tytonis	D	S	L	Dempster & Perrin (1990)
G. paeba	SU	L	S	
G. vallinus	D	S	L	Dempster & Perrin (1990)
Gerbillus paeba	SU	L	S	
Gerbillus pyramidum	D	L	L	Ziv *et al.* (1993)
G. allenbyi	SU	S	S	
Subfamily: Murinae				
Acomys cahirinus	D	L	S	Haim & Fluxman (1996)
A. russatus	SU	S	L	
Apodemus flavicollis	D	S	L	Hoffmeyer (1973), Montgomery (1978),
A. sylvaticus	SU	L	S	Cihákova & Frynta (1996)
A. flavicollis	D	S	L	Gliwicz (1981, 1984)
A. agrarius	SU	L	S	
Subfamily: Sigmodontinae				
Neotoma albigula	D	L	L	Dial (1988)
N. stephensi	SU	S	S	
N. albigula	D	L		Dial (1988)
N. devia	SU	S		
N. stephensi	D	L		Dial (1988)
N. devia	SU	S		
N. fuscipes	D	S	L	Cameron (1971)
N. lepida	SU	L	S	
Peromyscus leucopus	D	S	L	Fitch (1963), Kantak (1983, 1996), Long
P. maniculatus	SU	L	S	(1996); Disagrees: Brown (1964), Wolff *et al.* (1983)
P. leucopus	D	L	L	Brown (1964)
P. boylii	SU	S	S	
P. gossypinus	D	S	L	McCarley (1963)
P. leucopus	SU	L	S	
P. oreas	D	S	S	Sheppe (1967)
P. maniculatus	SU	L	L	
P. boylii	D	S	L	Holbrook (1979)
P. maniculatus	SU	L	S	
P. californicus	D	S	L	MacMillen (1964)
P. eremicus	SU	L	S	
P. californicus	D	S	L	MacMillen (1964)
P. maniculatus	SU	L	S	
Sigmodon fulviventer	D	S		Petersen (1973), Petersen & Helland (1978)
S. hispidus	SU	L		

* *Eutamias* in original source.
† *Perognathus* in original source.
‡ Misidentified as *M. arvalis* in Myllymäki (1977) (see Norrdahl & Korpimäki, 1993).

Appendix 2 Explanations of decisions made when studies disagreed

Peromyscus leucopus vs. P. maniculatus.
Laboratory experiments and field observations of Fitch (1963), Kantak (1983, 1996) and Long (1996) suggest that *P. leucopus* is dominant over *P. maniculatus*. These studies were given priority because other studies have found no advantage of one species over the other (e.g. Brown, 1964; Wolff *et al.*, 1983).

Sorex araneus vs. S. minutus.
The study of Dickman (1991) was given priority because he gave experimental evidence for currently acting competition, whereas Ellenbroek (1980) claimed that competition between these species had only occurred in the past.

Tamias amoenus vs. T. speciosus.
The field experiment of Chappell (1978) was given priority over the laboratory experiment of Heller (1971).

Thomomys bottae vs. T. talpoides.
The field data of Miller (1964) and Vaughan & Hansen (1964) were given greater weight than the laboratory data of Baker (1974).